6.1.1 INCREMENTED DATE

ALGORITHM

Step 1: Start

Step 2: Input day, month, year

Step 3: If year ≤ 0 → Print "Invalid Date" and Stop

Step 4: If month < 1 OR month > 12 → Print "Invalid Date" and Stop

Step 5: Determine number of days in the month:

If month = 1,3,5,7,8,10,12 → days = 31

If month = 4,6,9,11 → days = 30

If month = 2:

If leap year → days = 29

Else → days = 28

(Leap year condition:
year % 400 == 0 OR
(year % 4 == 0 AND year % 100 ≠ 0))

Step 6: If day < 1 OR day > days → Print "Invalid Date" and Stop

Step 7: If day < days

    day = day + 1

Step 8: Else (day == days)

day = 1

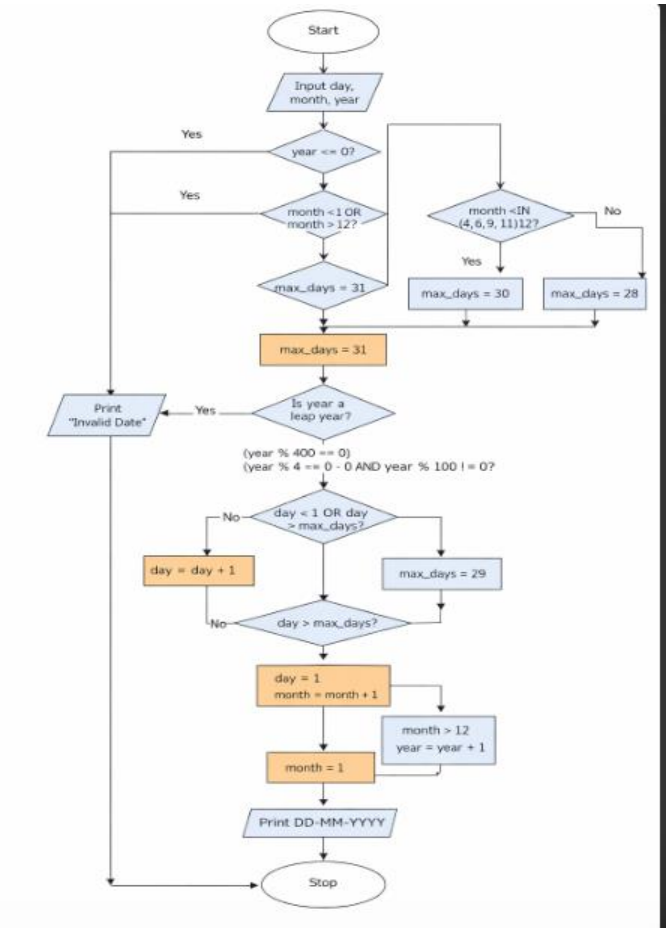
If month == 12
    month = 1
    year = year + 1

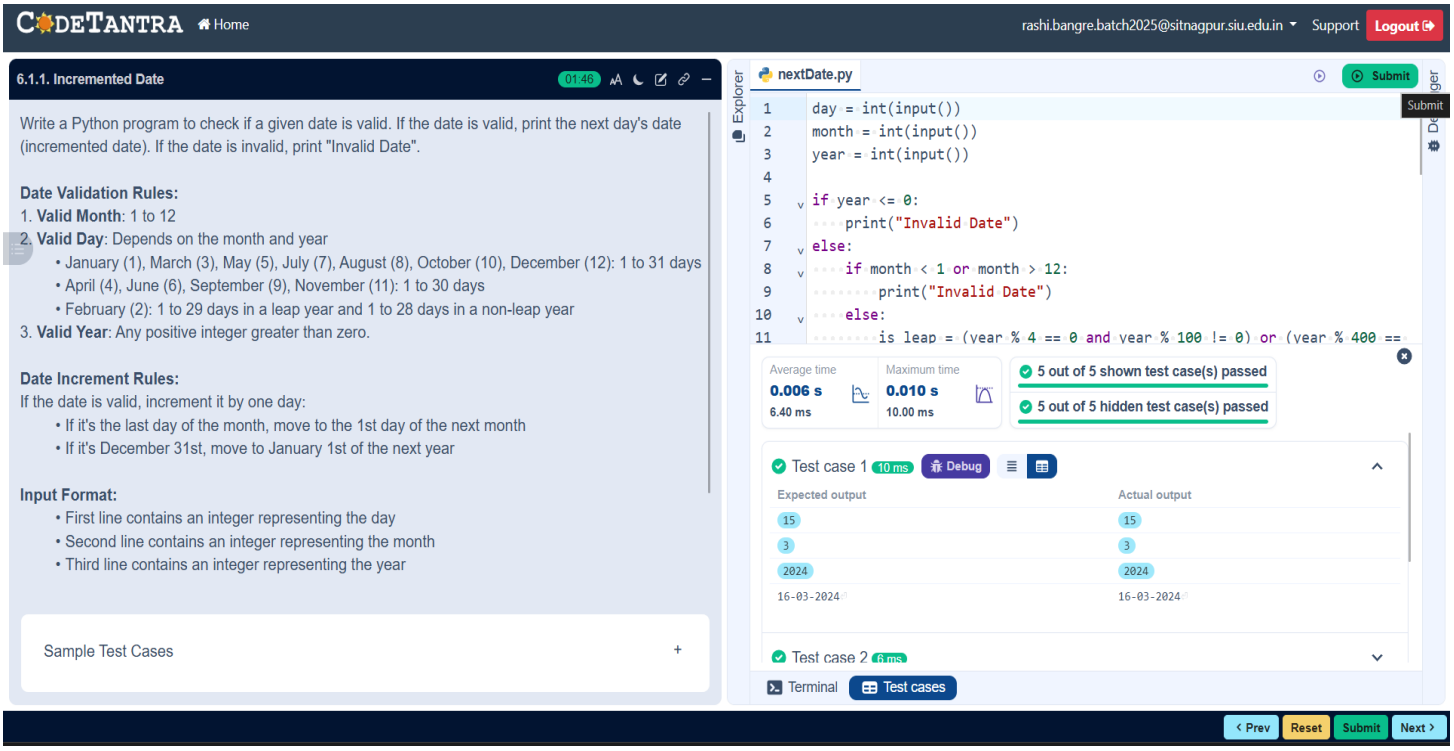
Else
    month = month + 1

Step 9: Print date in DD-MM-YYYY format

Step 10: Stop

CODETANTRA:-


Start
Input day, month, year
year <= 0?
Yes
month < 1 OR month > 12?
Yes
max_days = 31
month <IN (4, 6, 9, 11)12?
No
Yes
max_days = 30
max_days = 28
max_days = 31
Is year a leap year?
Yes
Print "Invalid Date"
(year % 400 == 0)
(year % 4 == 0 - 0 AND year % 100 != 0?
day < 1 OR day > max_days?
No
day = day + 1
max_days = 29
day > max_days?
No
day = 1
month = month + 1
month > 12
year = year + 1
month = 1
Print DD-MM-YYYY
Stop


CODETANTRA
Home
rashi.bangre.batch2025@sitnagpur.siu.edu.in
Support
Logout
6.1.1. Incremented Date
01:46
Write a Python program to check if a given date is valid. If the date is valid, print the next day's date (incremented date). If the date is invalid, print "Invalid Date".
Date Validation Rules:
1. Valid Month: 1 to 12
2. Valid Day: Depends on the month and year
• January (1), March (3), May (5), July (7), August (8), October (10), December (12): 1 to 31 days
• April (4), June (6), September (9), November (11): 1 to 30 days
• February (2): 1 to 29 days in a leap year and 1 to 28 days in a non-leap year
3. Valid Year: Any positive integer greater than zero.
Date Increment Rules:
If the date is valid, increment it by one day:
• If it's the last day of the month, move to the 1st day of the next month
• If it's December 31st, move to January 1st of the next year
Input Format:
• First line contains an integer representing the day
• Second line contains an integer representing the month
• Third line contains an integer representing the year
Sample Test Cases
nextDate.py
Submit
day = int(input())
month = int(input())
year = int(input())
if year <= 0:
print("Invalid Date")
else:
if month < 1 or month > 12:
print("Invalid Date")
else:
is_leap = (year % 4 == 0 and year % 100 != 0) or (year % 400 ==
Average time
0.006 s
6.40 ms
Maximum time
0.010 s
10.00 ms
5 out of 5 shown test case(s) passed
5 out of 5 hidden test case(s) passed
Test case 1
10 ms
Debug
Expected output
Actual output
15
3
2024
16-03-2024
Test case 2
6 ms
Terminal
Test cases
Prev
Reset
Submit
Next